

Penetration Testing

Nmap-Scan-Comparison

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Introduction

In this practical, an internal VirtualBox network's Metasploitable2/Ubuntu virtual machine was the target of four separate Nmap scans conducted from a Kali Linux virtual machine. Finding specific information about active hosts and their logical structures is the primary goal of this stage of the penetration testing lifecycle. These scans assist in locating open ports (communication channels), services, and firewall activity.

Network Configuration

A Bridged Network Adapter in VirtualBox was used to set up the virtual machines. This allows communication between the target machine (Metasploitable) and the Kali Linux machine (attacker) by connecting to the same local network.

Because DHCP is used to automatically allocate IP addresses to each virtual machine, different IP addresses may be assigned each time the machines are started. To ensure reliable results, the target machine's IP address was confirmed before to each scan.

Because of the successful connection between the machines made possible by this setup, penetration testing tasks like Nmap scans could be performed successfully.

Screenshots of the network adapter settings for both the Kali Linux and Metasploitable virtual machines are included to demonstrate that both systems were configured on the same network, allowing successful communication.

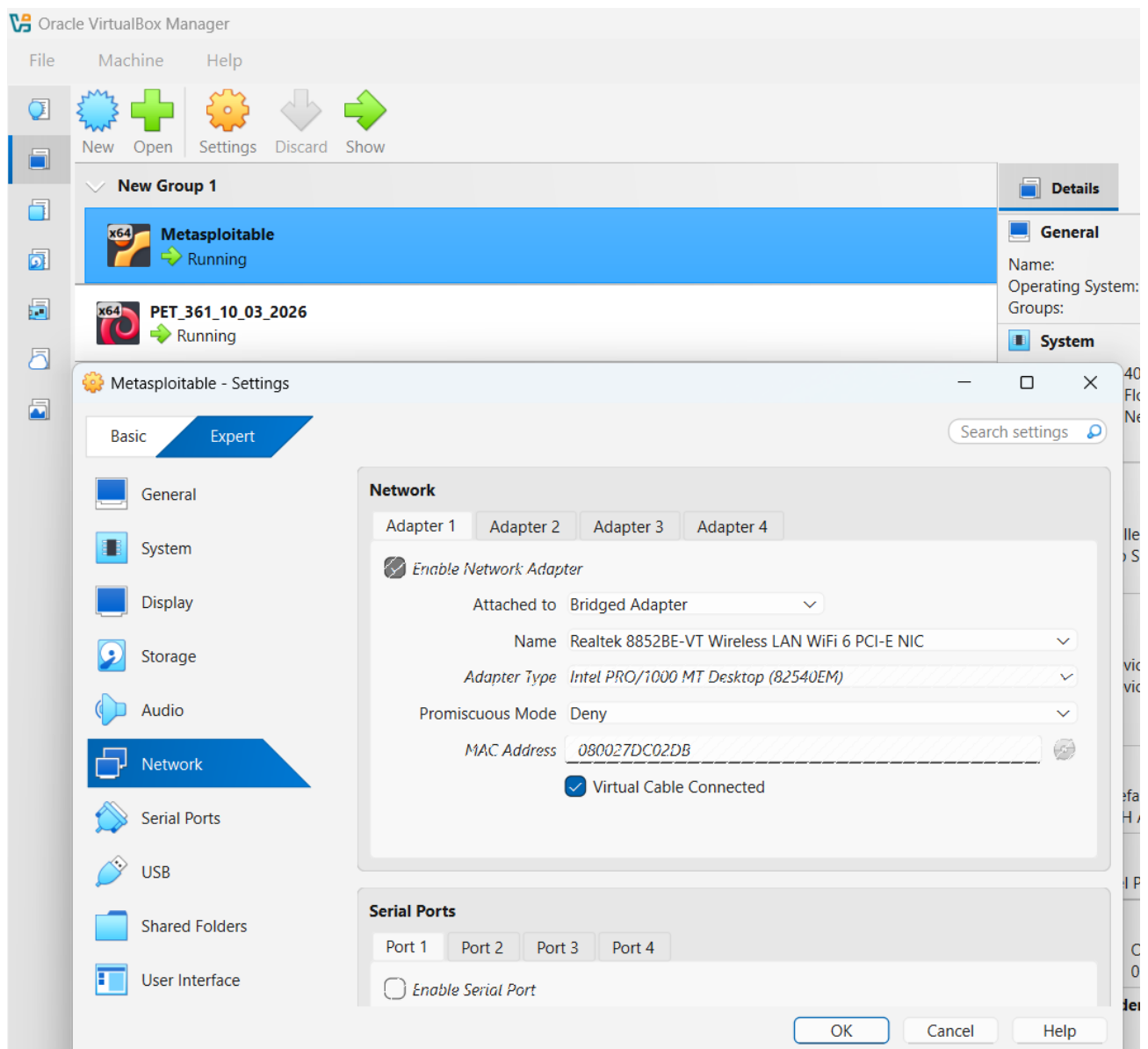


Figure 1: VirtualBox Network Adapter Configuration for Metasploitable

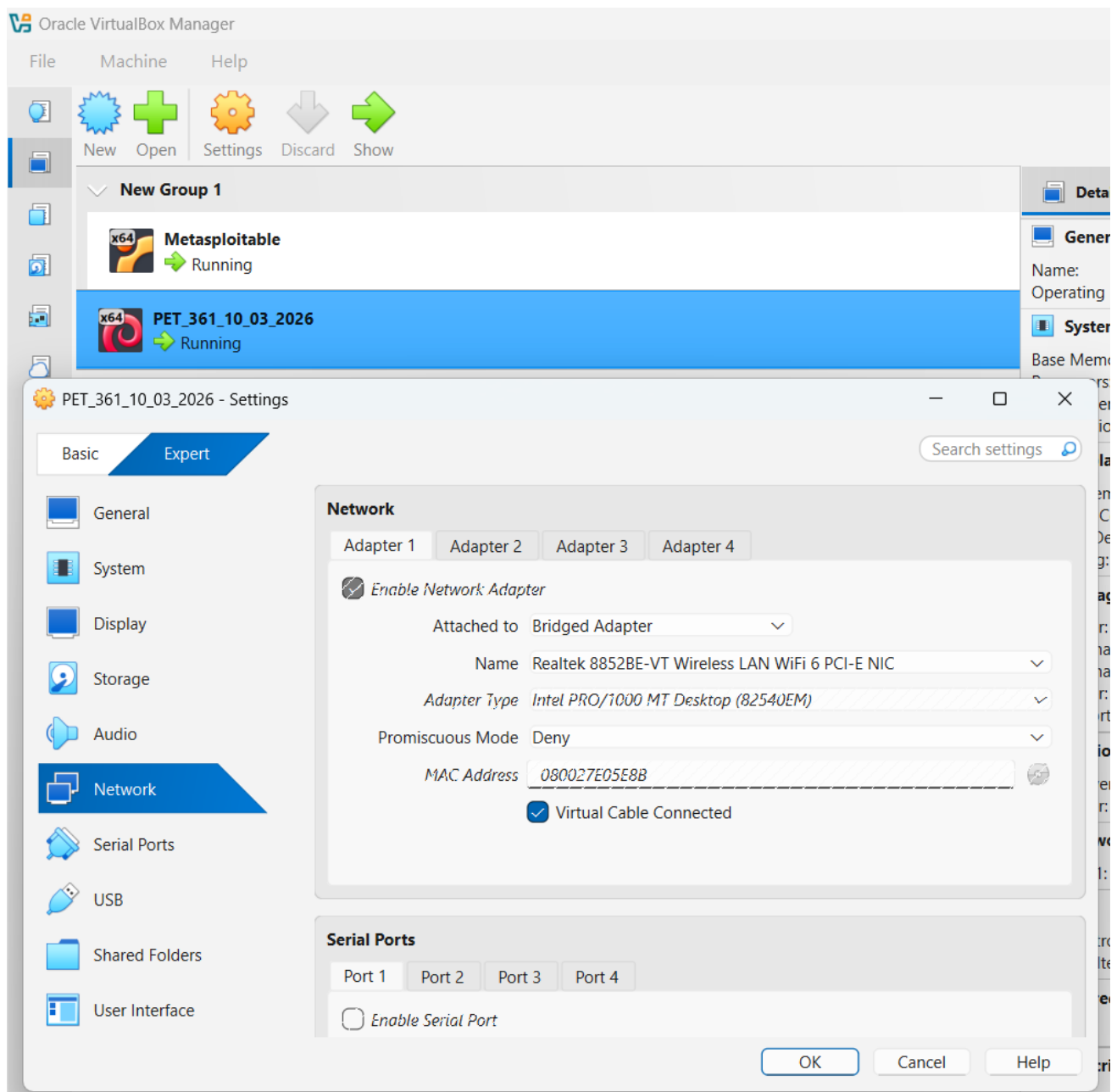
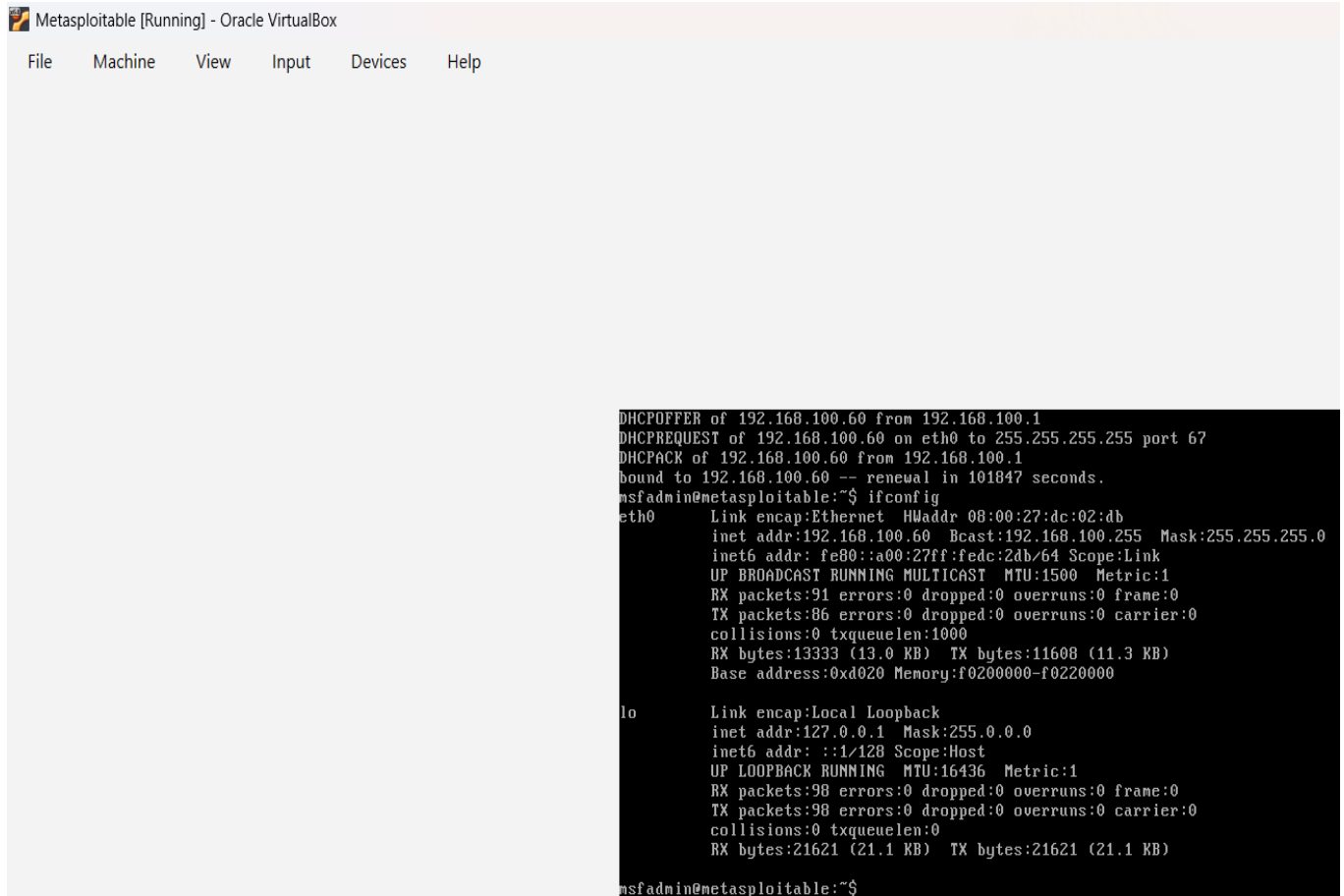


Figure 2: VirtualBox Network Adapter Configuration for Kali Linux

Network Setup and Connectivity Verification

IP Address Identification (Metasploitable)

A screenshot of a virtual machine window titled "Metasploitable [Running] - Oracle VirtualBox". The window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The main area is a terminal window showing the output of the `ifconfig` command. The output shows details for the `eth0` and `lo` interfaces. The `eth0` interface has an IP address of `192.168.100.60`. The `lo` interface has an IP address of `127.0.0.1`.

```
DHCPPOFFER of 192.168.100.60 from 192.168.100.1
DHCPREQUEST of 192.168.100.60 on eth0 to 255.255.255.255 port 67
DHCPACK of 192.168.100.60 from 192.168.100.1
bound to 192.168.100.60 -- renewal in 101847 seconds.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:dc:02:db
          inet addr:192.168.100.60  Bcast:192.168.100.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fedc:2db/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:91 errors:0 dropped:0 overruns:0 frame:0
          TX packets:86 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:13333 (13.0 KB)  TX bytes:11608 (11.3 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:98 errors:0 dropped:0 overruns:0 frame:0
          TX packets:98 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21621 (21.1 KB)  TX bytes:21621 (21.1 KB)

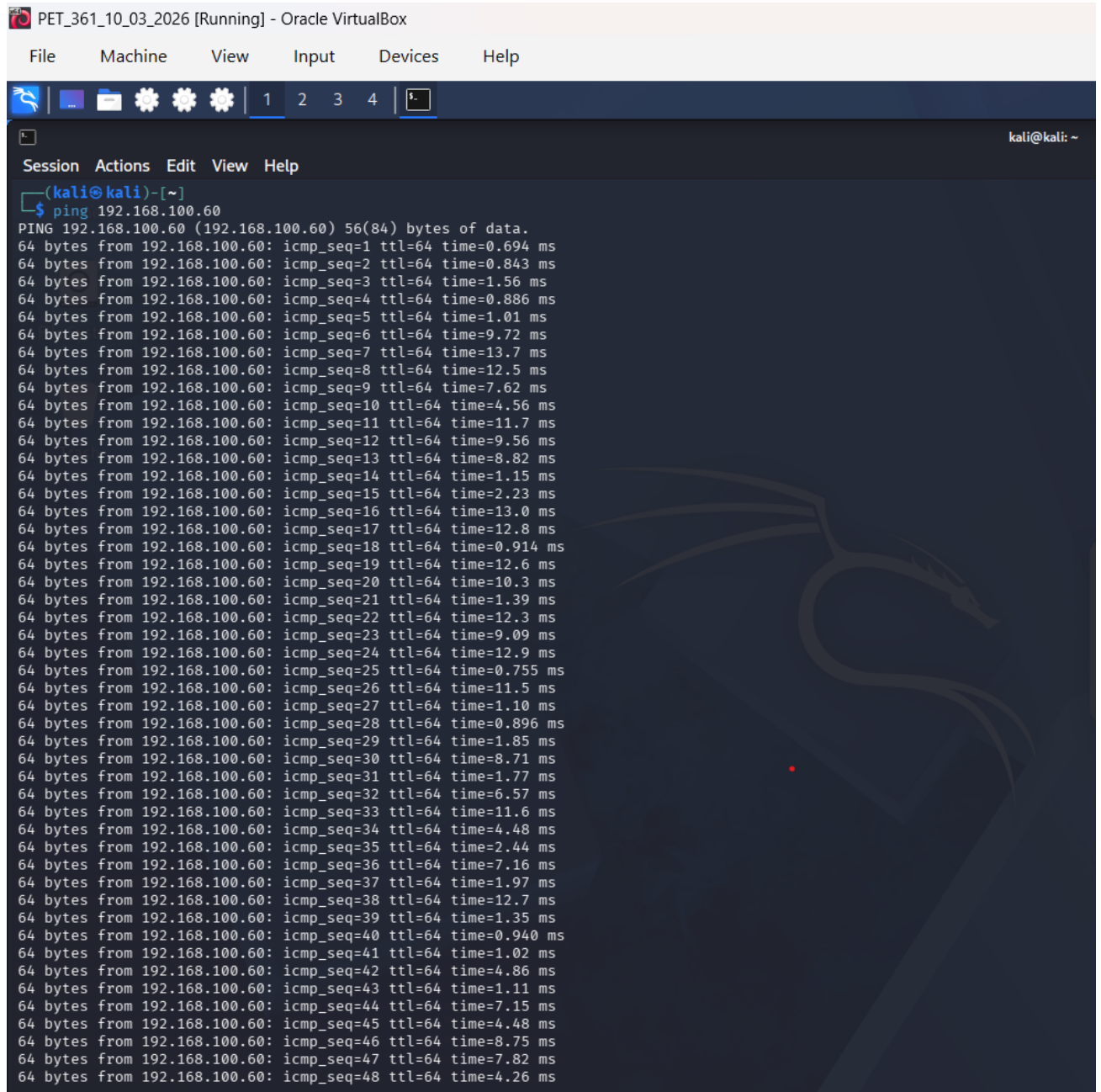
msfadmin@metasploitable:~$
```

Figure 3: IP Address Configuration on Metasploitable (ifconfig output)

The **ifconfig** command was used on the Metasploitable machine to find its IP address.

The IP address found was **192.168.100.60**, which was used as the target for all Nmap scans.

Connectivity Test (Ping Test)



The screenshot shows a terminal window titled "PET_361_10_03_2026 [Running] - Oracle VirtualBox". The terminal is running a ping command from a Kali Linux machine to the IP address 192.168.100.60. The output shows 48 successful replies, each 64 bytes in size, with varying round-trip times. A faint Kali Linux dragon logo is visible in the background of the terminal window.

```
PET_361_10_03_2026 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)~$ ping 192.168.100.60
PING 192.168.100.60 (192.168.100.60) 56(84) bytes of data.
64 bytes from 192.168.100.60: icmp_seq=1 ttl=64 time=0.694 ms
64 bytes from 192.168.100.60: icmp_seq=2 ttl=64 time=0.843 ms
64 bytes from 192.168.100.60: icmp_seq=3 ttl=64 time=1.56 ms
64 bytes from 192.168.100.60: icmp_seq=4 ttl=64 time=0.886 ms
64 bytes from 192.168.100.60: icmp_seq=5 ttl=64 time=1.01 ms
64 bytes from 192.168.100.60: icmp_seq=6 ttl=64 time=9.72 ms
64 bytes from 192.168.100.60: icmp_seq=7 ttl=64 time=13.7 ms
64 bytes from 192.168.100.60: icmp_seq=8 ttl=64 time=12.5 ms
64 bytes from 192.168.100.60: icmp_seq=9 ttl=64 time=7.62 ms
64 bytes from 192.168.100.60: icmp_seq=10 ttl=64 time=4.56 ms
64 bytes from 192.168.100.60: icmp_seq=11 ttl=64 time=11.7 ms
64 bytes from 192.168.100.60: icmp_seq=12 ttl=64 time=9.56 ms
64 bytes from 192.168.100.60: icmp_seq=13 ttl=64 time=8.82 ms
64 bytes from 192.168.100.60: icmp_seq=14 ttl=64 time=1.15 ms
64 bytes from 192.168.100.60: icmp_seq=15 ttl=64 time=2.23 ms
64 bytes from 192.168.100.60: icmp_seq=16 ttl=64 time=13.0 ms
64 bytes from 192.168.100.60: icmp_seq=17 ttl=64 time=12.8 ms
64 bytes from 192.168.100.60: icmp_seq=18 ttl=64 time=0.914 ms
64 bytes from 192.168.100.60: icmp_seq=19 ttl=64 time=12.6 ms
64 bytes from 192.168.100.60: icmp_seq=20 ttl=64 time=10.3 ms
64 bytes from 192.168.100.60: icmp_seq=21 ttl=64 time=1.39 ms
64 bytes from 192.168.100.60: icmp_seq=22 ttl=64 time=12.3 ms
64 bytes from 192.168.100.60: icmp_seq=23 ttl=64 time=9.09 ms
64 bytes from 192.168.100.60: icmp_seq=24 ttl=64 time=12.9 ms
64 bytes from 192.168.100.60: icmp_seq=25 ttl=64 time=0.755 ms
64 bytes from 192.168.100.60: icmp_seq=26 ttl=64 time=11.5 ms
64 bytes from 192.168.100.60: icmp_seq=27 ttl=64 time=1.10 ms
64 bytes from 192.168.100.60: icmp_seq=28 ttl=64 time=0.896 ms
64 bytes from 192.168.100.60: icmp_seq=29 ttl=64 time=1.85 ms
64 bytes from 192.168.100.60: icmp_seq=30 ttl=64 time=8.71 ms
64 bytes from 192.168.100.60: icmp_seq=31 ttl=64 time=1.77 ms
64 bytes from 192.168.100.60: icmp_seq=32 ttl=64 time=6.57 ms
64 bytes from 192.168.100.60: icmp_seq=33 ttl=64 time=11.6 ms
64 bytes from 192.168.100.60: icmp_seq=34 ttl=64 time=4.48 ms
64 bytes from 192.168.100.60: icmp_seq=35 ttl=64 time=2.44 ms
64 bytes from 192.168.100.60: icmp_seq=36 ttl=64 time=7.16 ms
64 bytes from 192.168.100.60: icmp_seq=37 ttl=64 time=1.97 ms
64 bytes from 192.168.100.60: icmp_seq=38 ttl=64 time=12.7 ms
64 bytes from 192.168.100.60: icmp_seq=39 ttl=64 time=1.35 ms
64 bytes from 192.168.100.60: icmp_seq=40 ttl=64 time=0.940 ms
64 bytes from 192.168.100.60: icmp_seq=41 ttl=64 time=1.02 ms
64 bytes from 192.168.100.60: icmp_seq=42 ttl=64 time=4.86 ms
64 bytes from 192.168.100.60: icmp_seq=43 ttl=64 time=1.11 ms
64 bytes from 192.168.100.60: icmp_seq=44 ttl=64 time=7.15 ms
64 bytes from 192.168.100.60: icmp_seq=45 ttl=64 time=4.48 ms
64 bytes from 192.168.100.60: icmp_seq=46 ttl=64 time=8.75 ms
64 bytes from 192.168.100.60: icmp_seq=47 ttl=64 time=7.82 ms
64 bytes from 192.168.100.60: icmp_seq=48 ttl=64 time=4.26 ms
```

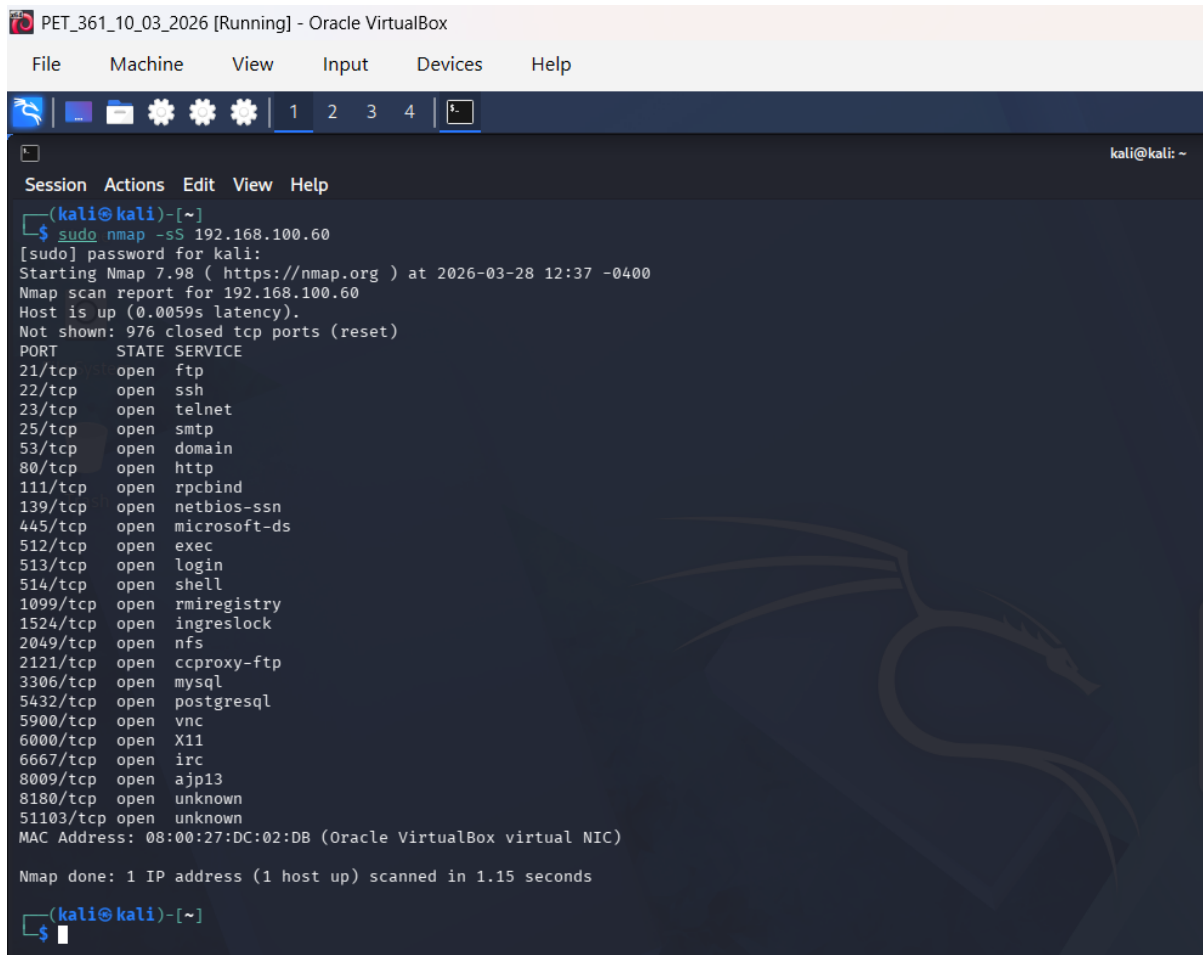
Figure 4: Ping Test Results Showing Connectivity Between Kali Linux and Target System

A ping test was performed from the Kali Linux machine to the target IP address (192.168.100.60) to verify network connectivity. The successful replies (64 bytes from...) confirm that the two machines can communicate over the network.

Explanation of Nmap Network Scanning

1. Stealth Scan (SYN Scan)

Command used: `sudo nmap -sS 192.168.100.60`

A screenshot of a terminal window titled 'PET_361_10_03_2026 [Running] - Oracle VirtualBox'. The terminal shows the execution of the command 'sudo nmap -sS 192.168.100.60'. The output indicates that Nmap 7.98 is starting at 2026-03-28 12:37 -0400. The scan report for 192.168.100.60 shows the host is up with a latency of 0.0059s. It lists 25 open ports with their corresponding services: 21/tcp (ftp), 22/tcp (ssh), 23/tcp (telnet), 25/tcp (smtp), 53/tcp (domain), 80/tcp (http), 111/tcp (rpcbind), 139/tcp (netbios-ssn), 445/tcp (microsoft-ds), 512/tcp (exec), 513/tcp (login), 514/tcp (shell), 1099/tcp (rmiregistry), 1524/tcp (ingreslock), 2049/tcp (nfs), 2121/tcp (ccproxy-ftp), 3306/tcp (mysql), 5432/tcp (postgresql), 5900/tcp (vnc), 6000/tcp (X11), 6667/tcp (irc), 8009/tcp (ajp13), 8180/tcp (unknown), and 51103/tcp (unknown). The MAC address is 08:00:27:DC:02:DB. The scan is completed in 1.15 seconds.

```
PET_361_10_03_2026 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~]
$ sudo nmap -sS 192.168.100.60
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-28 12:37 -0400
Nmap scan report for 192.168.100.60
Host is up (0.0059s latency).
Not shown: 976 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
51103/tcp open  unknown
MAC Address: 08:00:27:DC:02:DB (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.15 seconds

(kali@kali)-[~]
$
```

Figure 5: Nmap Stealth Scan (-sS) Results on Target System

The explanation:

The command `sudo nmap -sS 192.168.100.60` was used to execute a Stealth Scan (SYN scan). Because of its speed and low detectability, this kind of scan is often used to find open ports. Without completing the entire TCP three-way handshake, it works by delivering a TCP SYN packet to the target computer and evaluating the answer.

This method is called a half-open scan. An open port indicates that it is prepared to create a connection by responding with a SYN/ACK when a SYN packet is sent. Nmap

immediately sends a reset (RST) packet to end the connection rather than delivering an ACK to complete the handshake. This reduces the possibility that the target system may log the scan by preventing the session from being fully formed.

The SYN scan is often employed in penetration testing to successfully identify open ports while reducing detection because of this behaviour.

Delivered Output:

Based on the scan findings, the host at IP address 192.168.100.60 is reachable and active, with a duration of 0.0059 seconds. 976 of the 1000 ports that were checked are closed, although a few are open and hosting services.

The port number, protocol (TCP), state (open, closed, or filtered), and related service are displayed in a table form. This gives a full understanding of every service that is currently running on the target system.

The open ports identified include:

- **21/tcp (FTP)** – File Transfer Protocol service
- **22/tcp (SSH)** – Secure remote login
- **23/tcp (Telnet)** – Insecure remote access service
- **25/tcp (SMTP)** – Email service
- **53/tcp (DNS)** – Domain name resolution
- **80/tcp (HTTP)** – Web server
- **111/tcp (rpcbind)** -Remote procedure calls
- **139 & 445/tcp (SMB)** – File sharing services
- **3306/tcp (MySQL)** – Database service
- **5432/tcp (PostgreSQL)** – Database service
- **5900/tcp (VNC)** – Remote desktop access
- **6667/tcp (IRC)** – Chat service

Remote shell services, which are known to be vulnerable, are indicated by additional open ports like 512, 513, and 514. Ports like **8180 and 51103** are marked as unknown, meaning Nmap could not confidently identify the service running on those ports. The

system's attack surface is increased by the high number of open ports, which indicates that several services are actively operating on it.

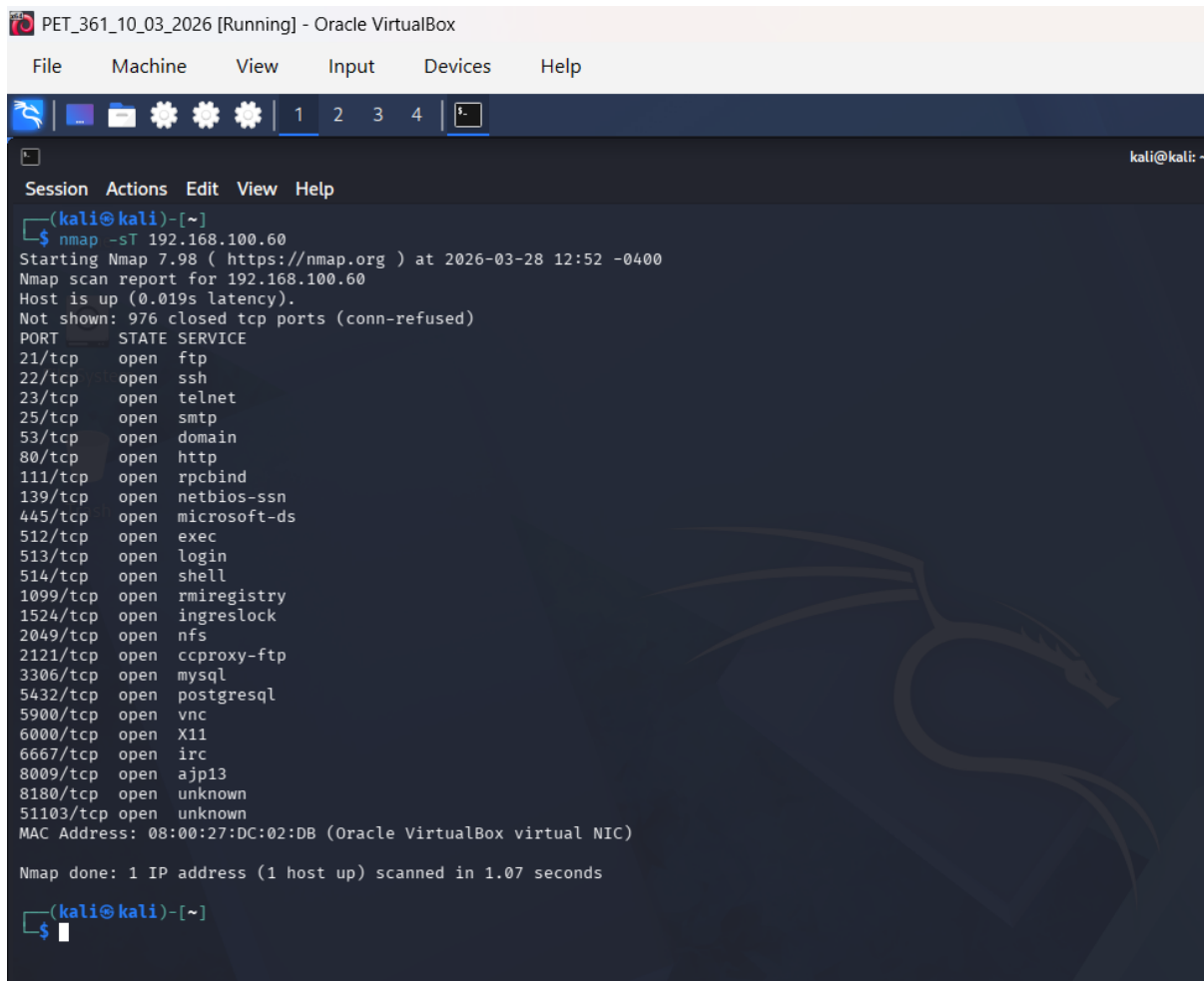
Considering everything, the scan offers a thorough list of all TCP services that are currently running on the target computer. Because the Metasploitable system is intentionally set up with problems for security testing, this is something to be expected.

The system's attack surface is increased by the numerous open ports, making it more vulnerable to exploitation.

(Parker, 2026)

2. TCP Connect Scan

Command: `nmap -sT 192.168.100.60`



```
PET_361_10_03_2026 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~]
$ nmap -sT 192.168.100.60
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-28 12:52 -0400
Nmap scan report for 192.168.100.60
Host is up (0.019s latency).
Not shown: 976 closed tcp ports (conn-refused)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
51103/tcp open  unknown
MAC Address: 08:00:27:DC:02:DB (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.07 seconds

(kali@kali)-[~]
$
```

Figure 6: Nmap TCP Connect Scan (-sT) Results on Target System

Explanation of the Scan:

The command `sudo nmap -sT 192.168.100.60` was used to do a TCP Connect Scan.

When a user has no administrator or raw packet privileges on the Kali Linux system, this scan is the default method. It creates a complete TCP three-way handshake with the target machine by using the network stack of the operating system.

Sending a SYN packet, getting a SYN/ACK response if the port is open, and then sending an ACK packet to finish the connection before ending it are all steps in the process. This method is very dependable in identifying whether ports are open because the entire connection is created.

The TCP Connect Scan, compared to the SYN (stealth) scan, completes the connection, which improves the target system's detection and logging of the activity. Because of this, even while it produces accurate findings, it is less discrete and more likely to be captured by security systems.

Delivered Output:

The host with IP address 192.168.100.60 is active and reachable, with a latency of 0.019 seconds, according to the scan findings. 976 of the 1000 ports that were checked are closed, although a few are open and hosting services.

The port number, protocol (TCP), status, and related service are displayed in a table manner. As is common with TCP Connect scans, the message "conn-refused" indicates that closed ports actively rejected the connection attempt.

The open ports identified include:

- **21/tcp (FTP)** – File Transfer Protocol service
- **22/tcp (SSH)** – Secure remote login
- **23/tcp (Telnet)** – Insecure remote access
- **25/tcp (SMTP)** – Email service
- **53/tcp (DNS)** – Domain name resolution
- **80/tcp (HTTP)** – Web server
- **139 & 445/tcp (SMB)** – File sharing services

- **3306/tcp (MySQL)** – Database service
- **5432/tcp (PostgreSQL)** – Database service
- **5900/tcp (VNC)** – Remote desktop access
- **6667/tcp (IRC)** – Chat service

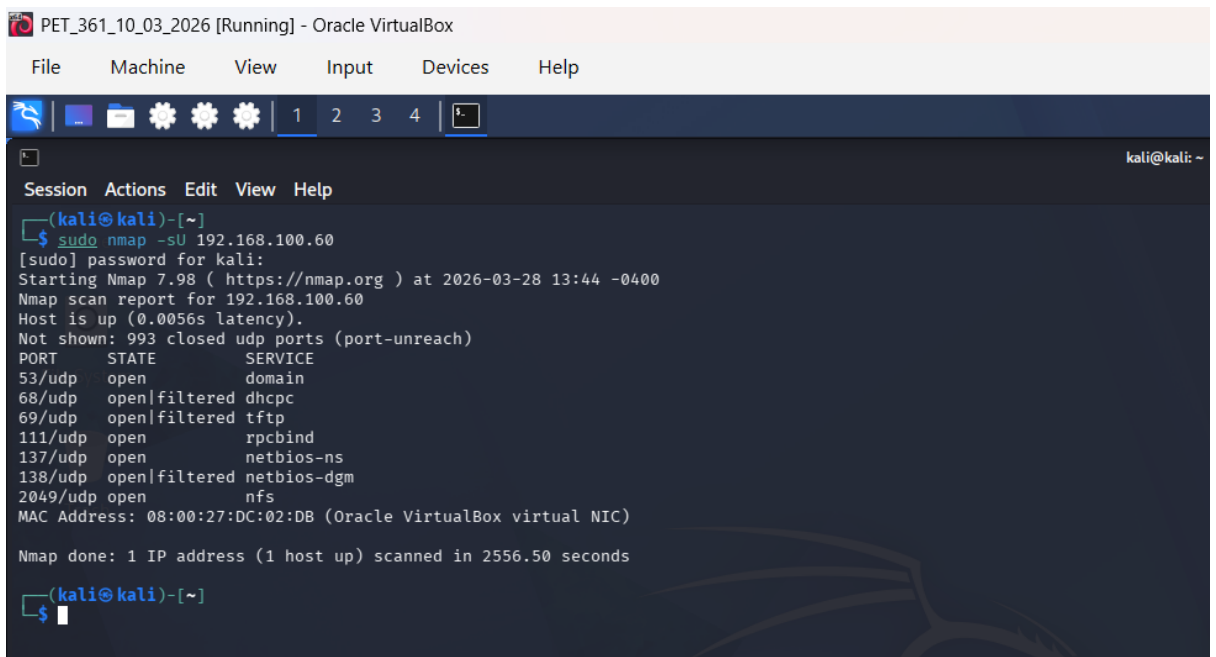
Remote shell services, which are known to be vulnerable, are indicated by additional ports like 512, 513, and 514. The system is more vulnerable to security threats when there are numerous open ports, which indicates that several services are operating on it.

Because both scans target the same ports and identify the same services, the output is quite similar to that of the SYN (stealth) scan. The main difference, however, is that the target system will log these connections since the TCP Connect Scan completely creates a connection. Therefore, the target system's service logs, including web server or authentication logs, may contain entries from the Kali Linux IP address.

(Cassella, 2023)

3. UDP Scan

Command: `sudo nmap -sU 192.168.100.60`



```
PET_361_10_03_2026 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~]
$ sudo nmap -sU 192.168.100.60
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-28 13:44 -0400
Nmap scan report for 192.168.100.60
Host is up (0.0056s latency).
Not shown: 993 closed udp ports (port-unreach)
PORT      STATE      SERVICE
53/udp    open       domain
68/udp    open|filtered dhcpd
69/udp    open|filtered tftp
111/udp   open       rpcbind
137/udp   open       netbios-ns
138/udp   open|filtered netbios-dgm
2049/udp  open       nfs
MAC Address: 08:00:27:DC:02:DB (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 2556.50 seconds
(kali@kali)-[~]
$
```

Figure 7: Nmap UDP Scan (-sU) Results on Target System

The explanation:

Services like DNS, DHCP, and SNMP that do not use the TCP protocol can be found using a UDP scan. As a connectionless protocol, UDP lacks an authentication procedure to confirm whether a port is actually "open." If Nmap receives no response after sending a raw UDP packet to the target, it proves that the port is either open or blocked by a firewall. The target machine will often return an ICMP "Port Unreachable" report if the port is closed. These scans are less certain and slower than TCP scans due to the lack of final confirmation.

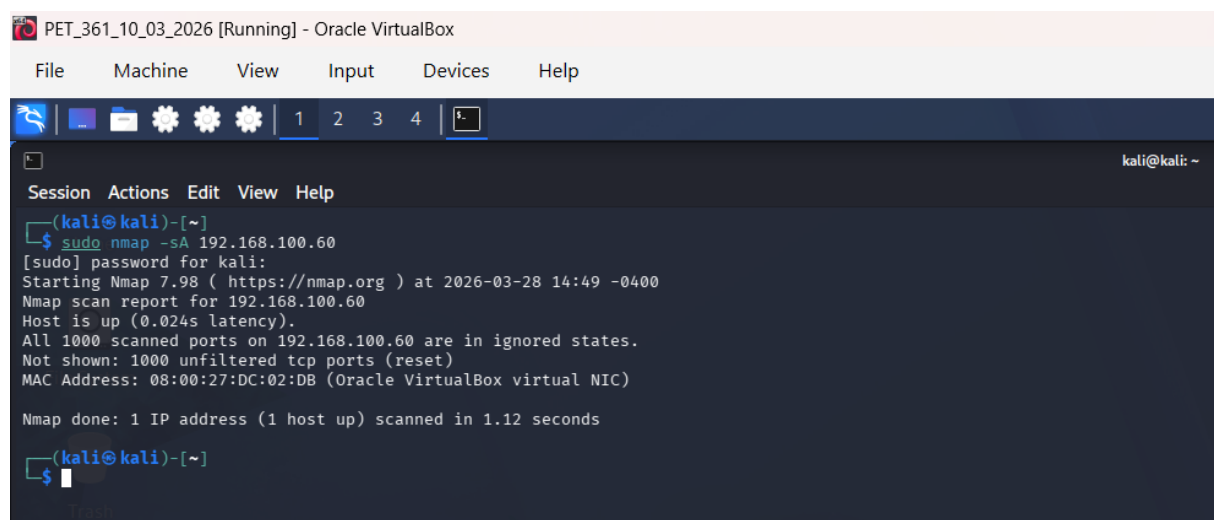
Delivered Output:

Ports are often labelled as open|filtered in the output. This shows that Nmap did not receive a response, making it impossible to identify with clarity if the service is running or if the packet was simply dropped. Common UDP services like port 53 (Domain) or 161 (SNMP) will be listed in this way on a Metasploit virtual machine.

(Code, 2023)

4. ACK Scan

Command: `sudo nmap -sA 192.168.100.60`



```
PET_361_10_03_2026 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~]
$ sudo nmap -sA 192.168.100.60
[sudo] password for kali:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-28 14:49 -0400
Nmap scan report for 192.168.100.60
Host is up (0.024s latency).
All 1000 scanned ports on 192.168.100.60 are in ignored states.
Not shown: 1000 unfiltered tcp ports (reset)
MAC Address: 08:00:27:DC:02:DB (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.12 seconds

(kali@kali)-[~]
$
```

Figure 8: Nmap ACK Scan (-sA) Results on Target System

The explanation:

The command `sudo nmap -sA 192.168.100.60` was used to conduct an ACK Scan. Instead of finding open ports or services, this scan is designed to ascertain whether ports are filtered or unfiltered by a firewall.

The ACK scan does not try to connect, in contrast to SYN or TCP Connect scans. Rather, it transmits TCP ACK packets to the target system as though they are a part of a session that has already been started.

The scan operates by examining the target system's response:

The port is categorised as unfiltered (reachable) if an RST (reset) packet is received, and as filtered (restricted by a security system) if there is no response or an ICMP error.

Because it is not intended to find open ports, the ACK scan differs from other scans. Rather, it is employed to map security system rules and identify the statelessness or statefulness of a security system. The target system will reply to unexpected ACK probes with RST packets if it is not shielded by a security system. However, the packets could be stopped or discarded if a firewall is in place.

As a result, rather than identifying active services, the ACK scan is helpful for discovering network filtering and comprehending the target system's security settings.

Delivered Output:

The host with IP address 192.168.100.60 is live and reachable, according to the scan findings.

The report shows that 1000 ports are categorised as unfiltered (reset), and all 1000 scanned ports are in ignored statuses. This indicates that the ports are reachable and not blocked by a firewall because the destination system returned RST answers to every ACK message.

Ports are not shown as "open" or "closed" by this scan. Rather, it categorises them as either unfiltered (reachable) or filtered (blocked). Since no filtering rules are used in this instance, all ports are unfiltered.

The results show that all 1000 scanned ports are **unfiltered**, meaning the target system responded with RST packets to all ACK probes. This indicates that the ports are

reachable and that no firewall is actively filtering the traffic. However, this scan does not determine whether ports are open or closed.

The ACK scan offers information on the network's security posture, particularly whether traffic is being restricted, to concentrating on the services that are operating on the system.

This behaviour suggests the absence of a stateful firewall, as such firewalls typically drop unsolicited ACK packets instead of responding.

Overall, the scan shows that there are no limitations preventing packets from accessing the system's ports and that it is completely reachable across the network.

(Unknown, 2025)

Comparison of Scan Results and Differences

The outcomes of the different Nmap scans demonstrate both similarities along with important distinctions in the ways that each scan method functions and the data it produces about the target system.

The reason the TCP Connect Scan (-sT) and SYN scan (-sS) are similar:

When it came to the open ports and services found, the Stealth Scan (-sS) and TCP Connect Scan (-sT) yielded identical results. This is because the goal of both scans is to find any open TCP ports on the target system.

The way the scans interact with the target is where the main distinction is found. The TCP Connect scan completes the entire three-way handshake, making it more dependable but also more visible in system logs, whereas the SYN scan employs a half-open connection, making it faster and less obvious.

Because the target system (Metasploitable) lacks advanced safety measures like firewalls and intrusion detection systems (IDS), both scans can identify the same services, which is why the results in this instance are so identical.

What makes the UDP Scan (-sU) different from the other scans is that:

Because it targets UDP services rather than TCP services, the UDP scan is quite different from TCP-based scans. Therefore, instead of identifying services like HTTP or SSH, it identifies an entirely separate set of ports, such as DNS (port 53) or SNMP (port 161).

Furthermore, there is no handshake to verify whether a port is open because UDP is a connectionless protocol. As a result, many results are tagged as open|filtered, indicating ambiguity.

Additionally, because many systems do not reply to UDP packets, the scan is slower and may result in warnings (such as retransmission limitations being surpassed). UDP scans become less dependable as a result, although they are still crucial for locating non-TCP services.

The reason the ACK scan is different:

When compared to the other scans, the ACK scan gives the most different outcomes. This is since its purpose is to look at firewall behaviour rather than identify open ports or services.

The ACK scan categorises ports as filtered or unfiltered rather than open or closed. In this instance, the target system reacted to every probe since every port was listed as unfiltered.

As would be expected in a Metasploitable environment, this shows that there is no active firewall filtering. Therefore, rather than the services that are operating on the system, the ACK scan offers information about the security setup of the network.

Differences in Table Form

Scan Type	Protocol	Handshake	Use Case	Goal	Expected Results on Metasploit
Stealth (-sS)	TCP	Partial (Half-open) SYN, SYN/ACK, RST	Speed and evasion.	Port discovery (Quiet)	Lists open TCP ports (21, 22, 80, etc.)
TCP Connect (-sT)	TCP	Full (3-way) SYN, SYN/ACK, ACK	Used when user lacks sudo/admin.	Port discovery (Noisy)	Identical port list to Stealth scan.
UDP (-sU)	UDP	None (Connectionless)	Finding non-TCP services.	Service discovery (DNS/DHCP)	Lists UDP ports (53, 67, 161).
ACK (-sA)	TCP	None (Probe only)	Firewall/ACL testing.	Firewall Mapping	Usually shows ports as “unfiltered.”

Conclusion

In conclusion, each form of Nmap scan has a specific function during a penetration test's reconnaissance stage. Although both TCP Connect and Stealth scans are important for identifying operational TCP services, their main differences are how they interact with the target's logs and how visible they are. While the ACK scan focuses on examining the target's firewall and defensive perimeter, the UDP scan widens this finding to connectionless services, but with less protection. We gain complete awareness of the attack surface and network security status of the target computer by applying all four techniques within the VirtualBox environment. This demonstrates the importance of using multiple scanning techniques to gain a complete understanding of both services and security controls.

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